```
d3.select("svg").style("background", "darkgray");
```

Infoviz is always cooler on a dark background.

NOTE The x-axis is drawn left-to-right, but the y-axis is drawn top-to-bottom, so you'll see that the circle is set 200 pixels to the right and 100 pixels down.

<canvas>

There's a second mode of drawing available with HTML5 using `<canvas>` elements to draw bitmaps. We won't go into detail here, but you'll see this method used in chapter 8 and again in chapter 11. The `<canvas>` element creates static graphics drawn in a manner similar to SVG that can then be saved as images. There are four main reasons to use `canvas`:

- *Compatibility*—We won't worry about this because if you're using D3, then you're coding for a modern browser.
- *Creating static images*—You can draw your data visualization with `canvas` to save views as snapshots for thumbnail and gallery views (this is what we'll do in chapter 8).
- *Large amounts of data*—SVG creates individual elements in the DOM, and although this is great for attaching events and styling, it can overwhelm a browser and cause significant slowdown (this is what we'll use `canvas` for in chapter 11).
- *WebGL*—The `<canvas>` element allows you to use WebGL to draw, so that you can create 3D objects. You can also create 3D objects like globes and polyhedrons using SVG, which we'll get into a bit in chapter 7 as we examine geospatial information visualization.

<CIRCLE>, <RECT>, <LINE>, <POLYGON>

SVG óprovides a set of common shapes, each of which has *attributes* that determine their size and position to make them easier to deal with than the generic `d` attribute you saw earlier. These attributes vary depending on the element that you're dealing with, so that `<rect>` has `x` and `y` attributes that determine the shape's top-left corner, as well as `height` and `width` attributes that determine its overall form. In comparison, the `<circle>` element has `cx` and `cy` attributes that determine the center of the circle, and an `r` attribute that determines the radius of the circle. The `<line>` element has `x1` and `y1` attributes that determine the starting point of the line and `x2` and `y2` attributes that determine its end point. There are other simple shapes that are similar to these, such as the `<ellipse>`, and there are more complex shapes, like the `<polygon>` with a `points` attribute that holds a set of comma-separated xy coordinates, in clockwise order, that determines the area bounded by the polygon.

Each of these attributes can be hand-edited in HTML to adjust its size, form, and position. Open up your element inspector, and click the `<rect>`. Change its `width` to 25 and its `height` to 25, as shown in figure 1.18.